PHILIP WOJCICKI

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EDUCATION

IMPERIAL COLLEGE LONDON '22 - '27
PhD in Reconfigurable Accelerators, advised by:

- Prof. Wayne Luk — Custom Computing
- Prof. Paul Kelly — Software Perf. Optimisation

Research on **FPGAs**, **AI acceleration**, **compilers**

IMPERIAL COLLEGE LONDON '18 - '22
MEng Computer Engineering (= *summa cum laude*):

- **Distinguished Final Year Project** (scored 88%)
- Y4: **Ranked #1 of 65, 1st Class Hons** (84%)
- Y3: **Ranked #4 of 65, 1st Class Hons** (80%)
- Y2: **Ranked #1 of 74, 1st Class Hons** (83%)
- Y1: **Ranked #1 of 74, 1st Class Hons** (80%)

RESEARCH EXPERIENCE

I have **presented my work** at:

- REACH'25 ScalPerf'25, ACACES'25, FDF'25, School of Microelectronics in Tianjin University, NANDA'23, CERN FastML workshops, FPT'22,

I have served as a **reviewer** for:

- FPGA, FCCM, CF, IMPACT, PACT, Euro-Par

TEACHING EXPERIENCE

Since 2020, I have been a **teaching assistant**:

- I develop and maintain labs and **coursework** about creating a C90 to RISC-V compiler in C++.
- That includes automation for testing, benchmarking, reviewing and environment deployment, with experience report planned for SIGCSE TS'27.
- I provide support during lab sessions, mark coursework, offer 1-to-1 tutorials. I have been awarded **Top Undergraduate Teaching Assistant** prize.

SKILLS

PROGRAMMING

- **Python, C++, SystemVerilog, RISC-V ASM**

TOOLS & TECHNOLOGIES

- ML frameworks (**PyTorch**, **HLS4ML**)
- Quant. Aware Training (**Brevitas**, **QPyTorch**)
- Hardware design/verification (**Vivado**, **Quartus**)
- Compiler infrastructure (**MLIR**, **LLVM**)
- Linux environment, version control (**Git**, **Bash**)
- Software profiling (**Intel VTune Profiler**, **perf**)

INDUSTRIAL EXPERIENCE

FRACTILE Nov '24 – Jan '25
Low-latency Researcher

- Evaluated latency and area trade-offs of block floating-point formats (**Microscaling**) for an analog in-memory-compute AI-accelerator ASIC.
- Implemented and benchmarked low-latency paths as **ASM RISC-V (RVV)** kernels and **SystemVerilog** blocks to guide HW/SW partitioning.

JUMP TRADING Sep '20 – Oct '20, Mar '21 – Sep '21
Software & Hardware Engineer

- Created a library in **Python** for exploring topology, configuring, connecting and graphing multi-device (ASIC, FPGA, CPU, GPU) hardware systems
- Built an efficient, distributed arbitrage trading **C++** application composed of several parallel processes running on ASICs, FPGAs, and x86 machines
- Configured and benchmarked formal verification using **SystemVerilog** and a novel custom **Python** tool for use in ASIC and FPGA development
- Built an ultra low latency ASIC validation test platform for floating point calculations on x86 and RISC-V architectures using **C++**, **C** and **Python**

ARM Jun '20 – Sep '20
Systems Architect

- Improved autonomous driving platform's verification in **SystemVerilog**, overhauled its documentation, added support for formal (**JasperGold**)

RESEARCH PROJECTS

MX ATTENTION ON FPGA [FCCM, code]

- Built a parametrised **SystemVerilog** FPGA attention architecture with operator-wise MX-style BFP formats and configurable accumulation.
- Developed a **Python** DSE framework, finding designs with **29% fewer LUTs**, **7% fewer FFs**, and **0.42 lower perplexity** than baseline.

QUANTISED TNN ON FPGA [FPT, code]

- Developed a novel TNN architecture for GPUs (PyTorch) and FPGAs (HLS) for High Energy Physics experiments in collaboration with CERN
- FPGA solution outperformed SoTA models on GPU by ∼**1000 times** thanks to software/hardware-aware optimisations, without accuracy loss
- Experimented with quantisation-aware training (QAT), developed a quick FPGA-friendly post-training quantisation (PTQ) scheme for HLS4ML

QUBIT VISION TNN ON FPGA [IEEE TQE - pending]

- Designed a low-latency FPGA pipeline (CameraLink) for real-time qubit-state classification in collaboration with experimental quantum physicists
- Optimised the ViT architecture to achieve millisecond-scale end-to-end detection with up to ∼**120**× **lower latency** than a GPU baseline

ZERO-SHOT LEARNING [ACM TECS - pending]

- Developed a lightweight zero-shot learning framework with attribute knowledge graphs, reducing parameters by ∼**100**× while retaining accuracy
- Designed an accelerator on FPGA for CNN feature extraction and attribute recognition, achieving ∼**67**× speedup over a software-only baseline

SYSTOLIC ARRAY COMPILER [ICCD - pending, code]

- Developed an MLIR-based HLS framework for **C/C++** and **PyTorch** programs to systolic-array accelerators using polyhedral optimisations.
- Achieved SoTA performance while enabling **MLIR** pass interchangeability.